



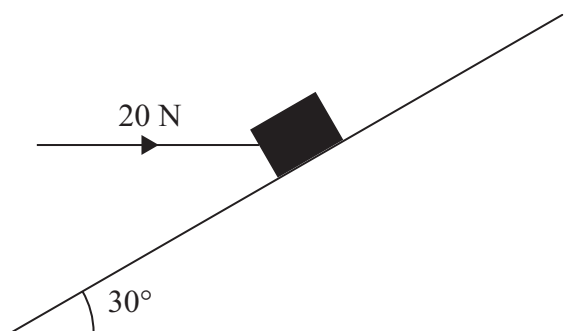
(3)

(b) Find the value of  $m$ .

(3)



3.



### Figure 2

A box of mass 5 kg lies on a rough plane inclined at  $30^\circ$  to the horizontal. The box is held in equilibrium by a horizontal force of magnitude 20 N, as shown in Figure 2. The force acts in a vertical plane containing a line of greatest slope of the inclined plane. The box is in equilibrium and on the point of moving down the plane. The box is modelled as a particle.

Find

- (a) the magnitude of the normal reaction of the plane on the box, (4)
- (b) the coefficient of friction between the box and the plane. (5)



4. A car is moving on a straight horizontal road. At time  $t = 0$ , the car is moving with speed  $20 \text{ m s}^{-1}$  and is at the point  $A$ . The car maintains the speed of  $20 \text{ m s}^{-1}$  for 25 s. The car then moves with constant deceleration  $0.4 \text{ m s}^{-2}$ , reducing its speed from  $20 \text{ m s}^{-1}$  to  $8 \text{ m s}^{-1}$ . The car then moves with constant speed  $8 \text{ m s}^{-1}$  for 60 s. The car then moves with constant acceleration until it is moving with speed  $20 \text{ m s}^{-1}$  at the point  $B$ .

(a) Sketch a speed-time graph to represent the motion of the car from  $A$  to  $B$ .

(3)

(b) Find the time for which the car is decelerating.

(2)

Given that the distance from  $A$  to  $B$  is 1960 m,

(c) find the time taken for the car to move from  $A$  to  $B$ .

(8)



(3)

(5)

(4)

(a) Find the direction in which  $S$  is moving, giving your answer as a bearing. (3)

(b) Write down  $\mathbf{s}$  in terms of  $t$ . (2)

(c) Find the distance of  $S$  from  $B$  when  $t = 3$  (4)

(d) Find the distance of  $S$  from  $B$  when  $S$  is due north of  $B$ . (4)

7.



### Figure 3

Two particles  $P$  and  $Q$ , of mass  $0.3\text{ kg}$  and  $0.5\text{ kg}$  respectively, are joined by a light horizontal rod. The system of the particles and the rod is at rest on a horizontal plane.

At time  $t = 0$ , a constant force  $\mathbf{F}$  of magnitude 4 N is applied to  $Q$  in the direction  $PQ$ , as shown in Figure 3. The system moves under the action of this force until  $t = 6$  s. During the motion, the resistance to the motion of  $P$  has constant magnitude 1 N and the resistance to the motion of  $Q$  has constant magnitude 2 N.

Find

- (a) the acceleration of the particles as the system moves under the action of  $\mathbf{F}$ ,

- (b) the speed of the particles at  $t = 6$  s,
- (2)**

- (c) the tension in the rod as the system moves under the action of  $\mathbf{F}$ . (3)

At  $t = 6$  s,  $\mathbf{F}$  is removed and the system decelerates to rest. The resistances to motion are unchanged. Find

- (d) the distance moved by  $P$  as the system decelerates,
- (4)**

- (e) the thrust in the rod as the system decelerates. (3)



